CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 16: Loci (Locus and its Constructions)

Previous year board questions, topic-wise: 5 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Loci in a triangle

1. Statement 1: The point equidistant from three non-collinear points D, E and F is the circumcentre of $\triangle DEF$. Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true ICSE 2025, 1 mark

2. The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes ICSE 2024, 1 mark

2 Construction questions

3. Using ruler and compasses, construct $\angle ABC = 90^\circ$ with $AB = 6$ cm and $BC = 8$ cm. Construct (a) the locus of points equidistant from B and C (b) the locus of points equidistant from A and B (c) the locus of points at the fixed distance OA from O, where O is the point common to (a) and (b) (d) the locus of points equidistant from BA and BC. ICSE 2024, 4 marks

4. Using ruler and compasses, draw a circle of radius 4.5 cm and a chord $AB = 6$ cm. (a) Construct the locus of points equidistant from A and B, and mark the point D where it meets the circle. (b) Join AD, construct the locus of points equidistant from AD and AB, and mark the point C where it meets the circle. (c) Join BC and CD and measure CD. ICSE 2020, 4 marks

5. Using ruler and compasses, construct a semicircle on the diameter $BC = 7$ cm. Locate the point A on the semicircle that is equidistant from B and C. Complete the cyclic quadrilateral $ABCD$ with D equidistant from AB and BC. Measure $\angle ADC$. ICSE 2019, 4 marks

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